Solve for u.

$$14 + 8(u + 12) = 2(6u + 19)$$

$u =$ [ ]

Save answer

Solve for c.

$$18 = 4(c - 13) + 2$$

$c =$ [ ]

Save answer

Solve for y.

$$8y = 2 + 6(y + 5)$$

$y =$ [ ]

Save answer

Solve for c.

$4c - 3c = 14$

$c =$

Save answer

Solve for b.

$9 = -7 + 2(1 + b)$

$b =$

Save answer

Solve for h.

$-10 = h + 5(h + 10)$

$h =$

Save answer

Solve for w.

$$-3w = 9 - 6w$$

$w =$ [ ]

Save answer

Solve for u.

$$10u = 7 + 9u$$

$u =$ [ ]

Save answer

Solve for b.

$$10 + 4b = 6b - 10$$

$b =$ [ ]

Save answer

Solve for q.

$$7q - 5q = 16$$

$q =$ [ ]

Save answer

Solve for j.

$$7j = 8j - 3$$

$j =$ [ ]

Save answer

Solve for c.

$$c = -9 - 8c$$

$c =$ [ ]

Save answer

Solve for g.

$$16g - 11g + g = 6$$

$g =$ [ ]

Save answer

Solve for u.

$$-9 - 4u = -u$$

$u =$ [ ]

Save answer

Solve for q.

$$10 - 5q = -3q$$

$q =$ [ ]

Save answer

Find the slope of the line that passes through (10, 7) and (1, 12).

Simplify your answer and write it as a proper fraction, improper fraction, or integer.

Save answer

Skip to next

A line passes through the points in this table.

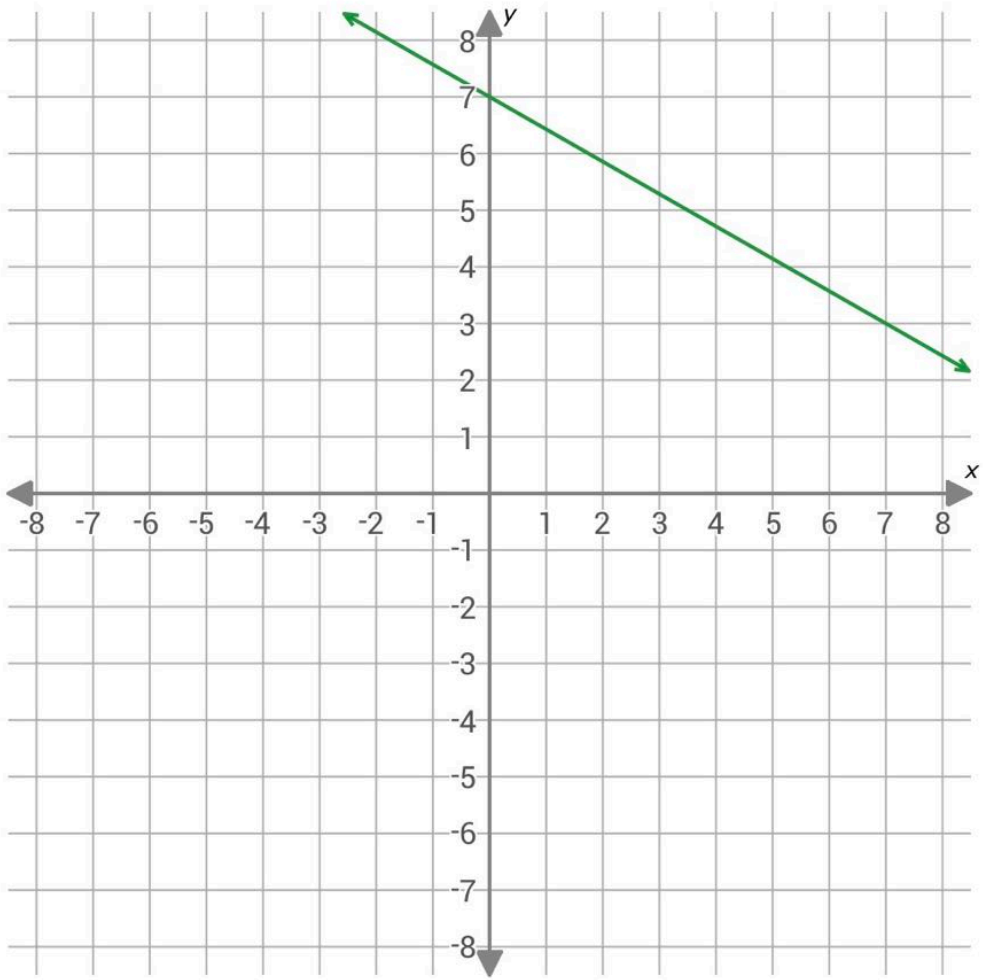
x	y
-4	12
-2	0
-1	-6
3	-30

What is the slope of the line?

Write your answer as an integer or simplified fraction.

Save answer

Write the equation of this line in slope-intercept form.



Write your answer using integers, proper fractions, and improper fractions in simplest form.

A line has a slope of 8 and a y-intercept of -5. Write its equation in slope-intercept form.

Write your answer using integers, proper fractions, and improper fractions in simplest form.

Save answer

Skip to next

Find the slope of the line that passes through (6, 16) and (10, 9).

Simplify your answer and write it as a proper fraction, improper fraction, or integer.

Save answer

Skip to next

Find the slope of the line that passes through (10, 7) and (1, 12).

Simplify your answer and write it as a proper fraction, improper fraction, or integer.

Save answer

Find the y-intercept of the line $y = -3x - 9$.

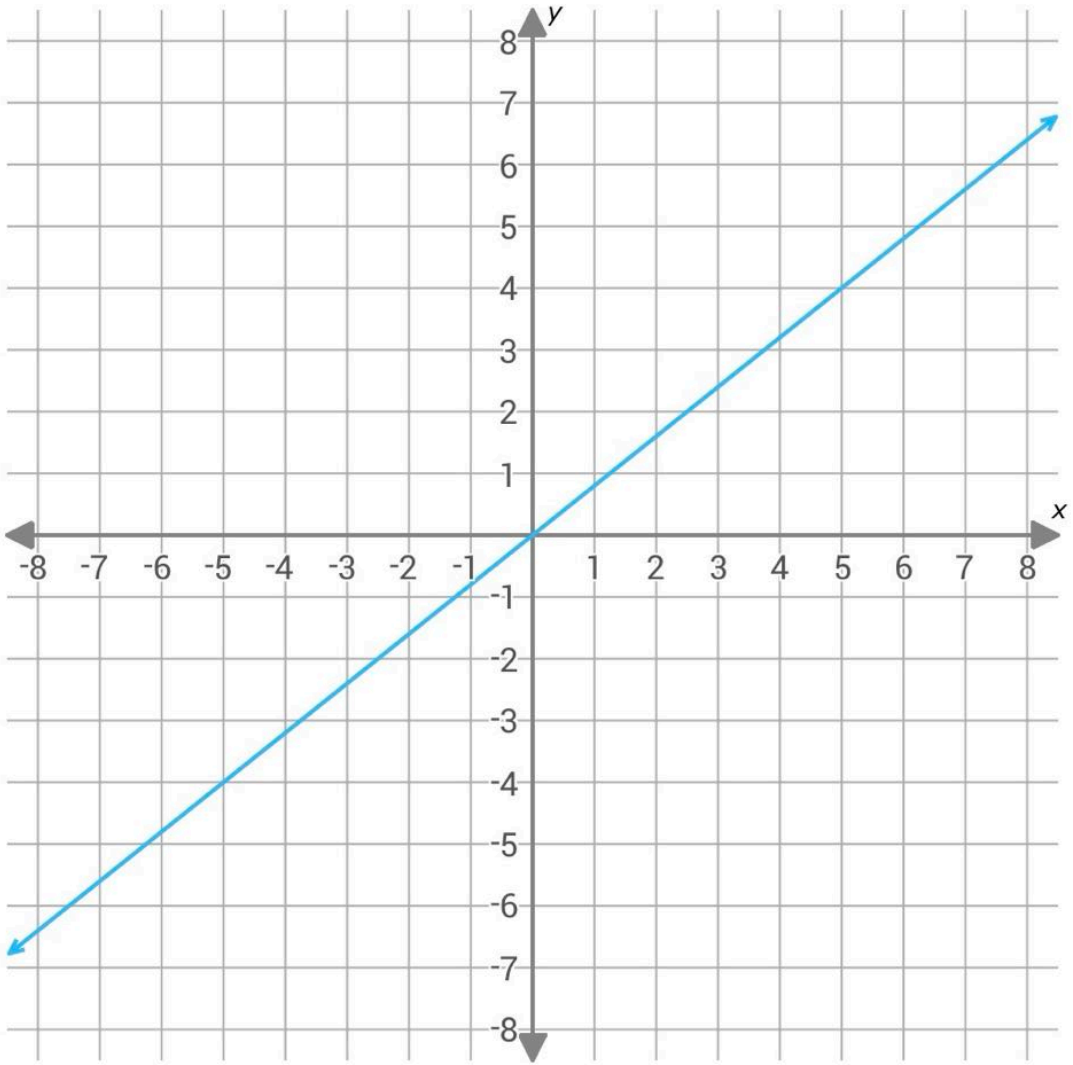
Write your answer as an integer or as a simplified proper or improper fraction, not as an ordered pair.

Save answer



Skip to next question >

What is the slope of this line?



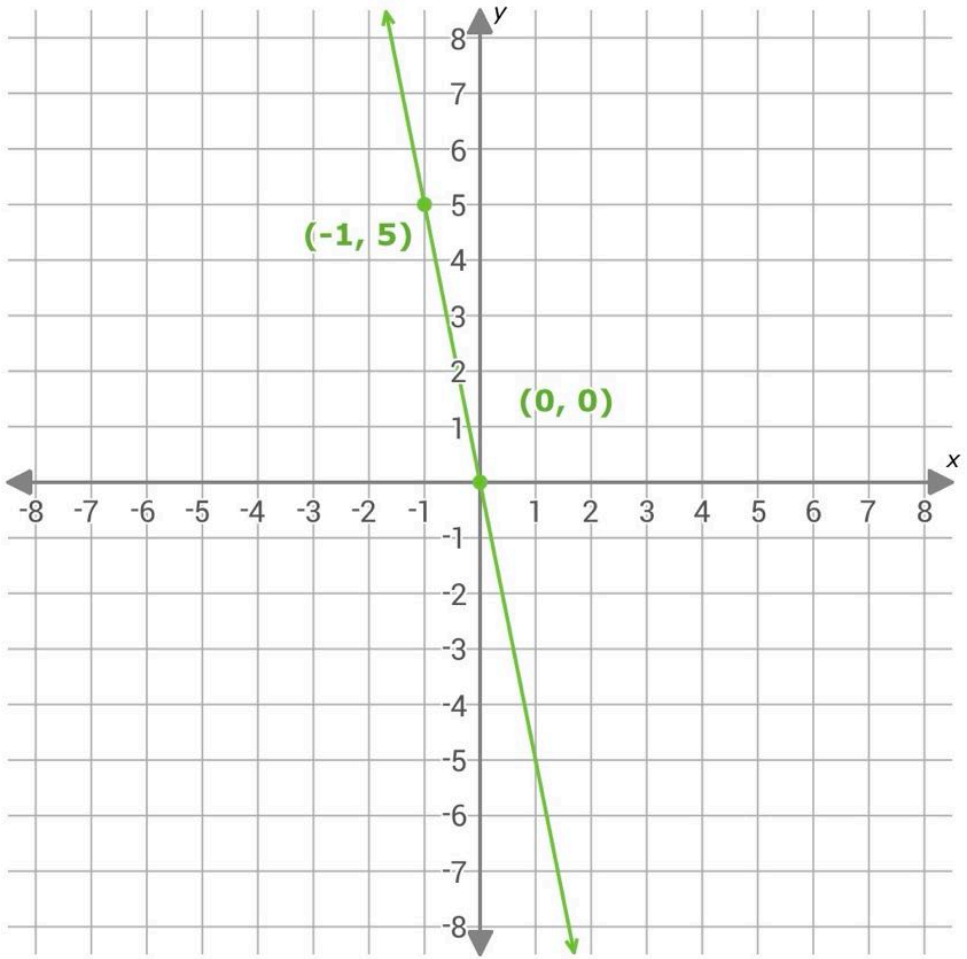
Simplify your answer and write it as a proper fraction, improper fraction, or integer.

Find the slope of the line that passes through (10, 10) and (2, 1).

Simplify your answer and write it as a proper fraction, improper fraction, or integer.

Save answer

What is the slope of this line?



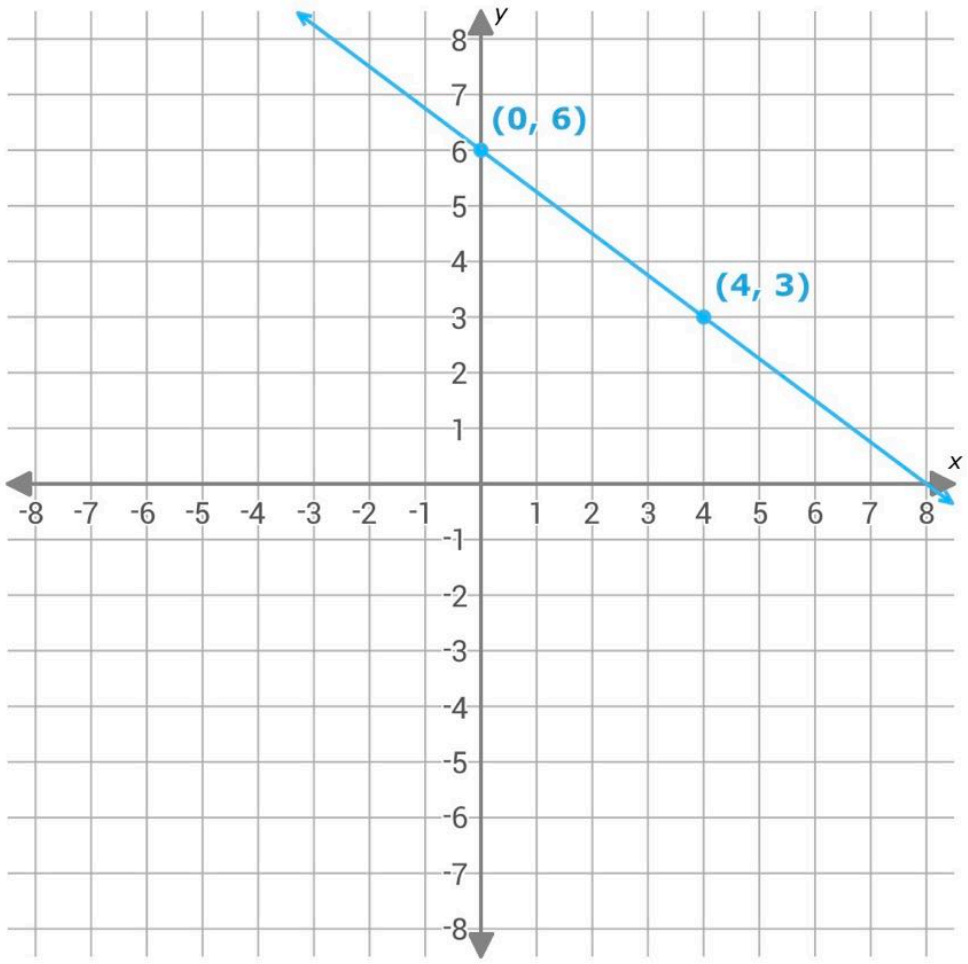
Simplify your answer and write it as a proper fraction, improper fraction, or integer.

Find the y-intercept of the line $y = \frac{7}{9}x + \frac{2}{3}$.

Write your answer as an integer or as a simplified proper or improper fraction, not as an ordered pair.

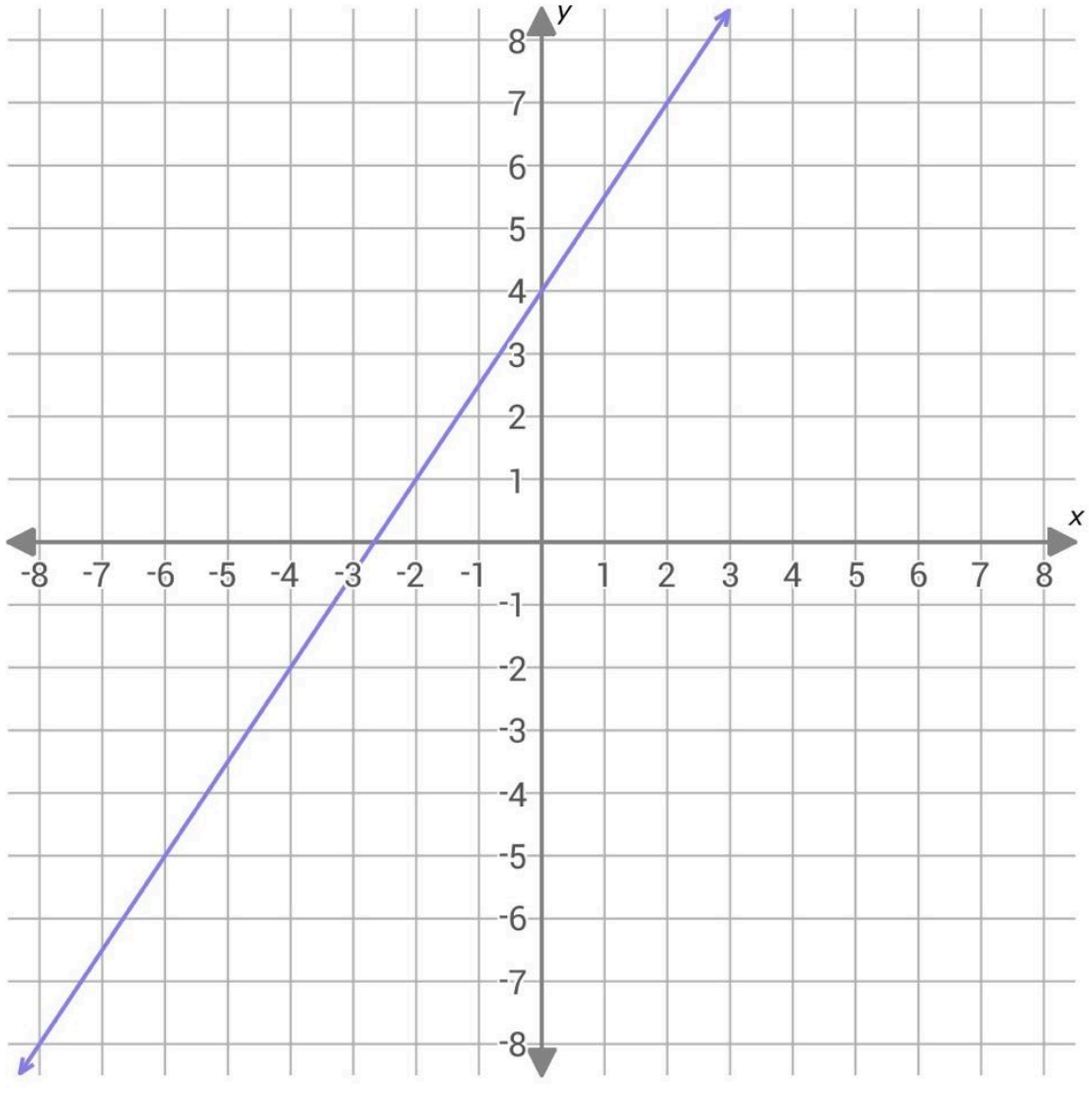
Save answer

What is the slope of this line?



Simplify your answer and write it as a proper fraction, improper fraction, or integer.

What is the slope of this line?



Simplify your answer and write it as a proper fraction, improper fraction, or integer.

Find the slope of the line that passes through (9, 10) and (3, 5).

Simplify your answer and write it as a proper fraction, improper fraction, or integer.

Save answer

Find the y-intercept of the line $y = 6x + 8$.

Write your answer as an integer or as a simplified proper or improper fraction, not as an ordered pair.

Save answer



Skip to next question >

Find the slope of the line $y = \frac{1}{6}x + \frac{4}{11}$.

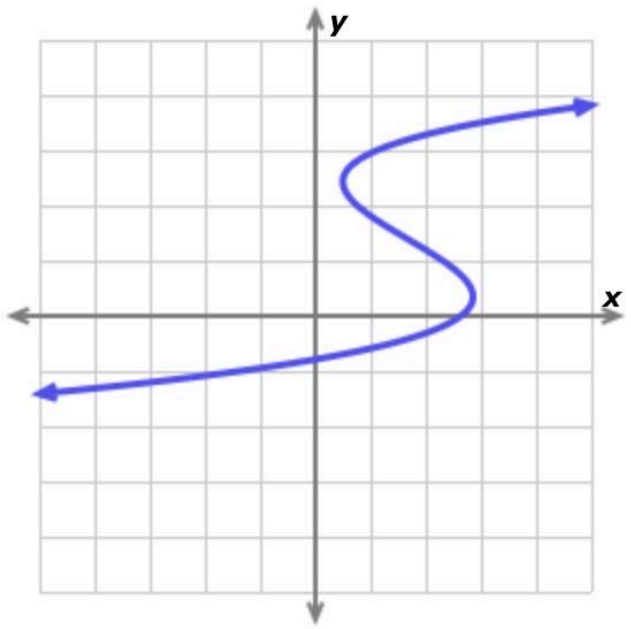
Write your answer as an integer or as a simplified proper or improper fraction.

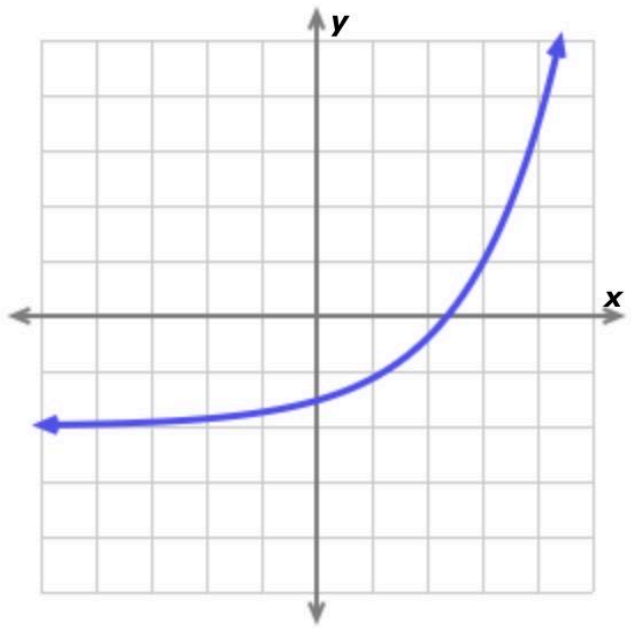
Save answer

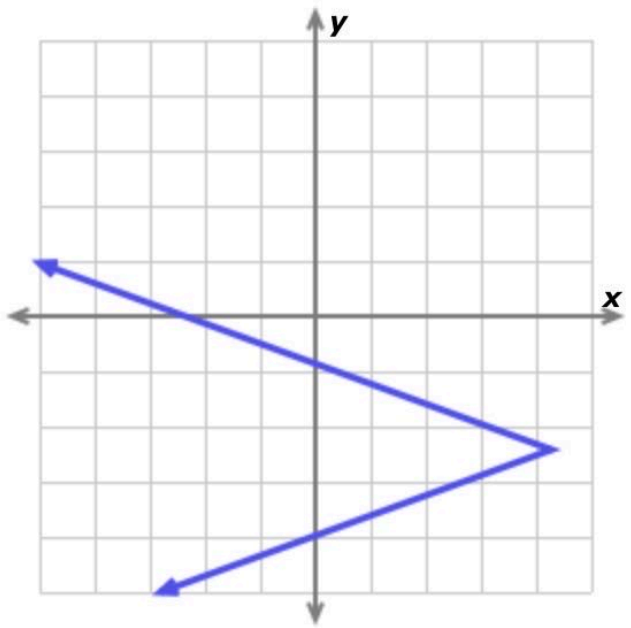


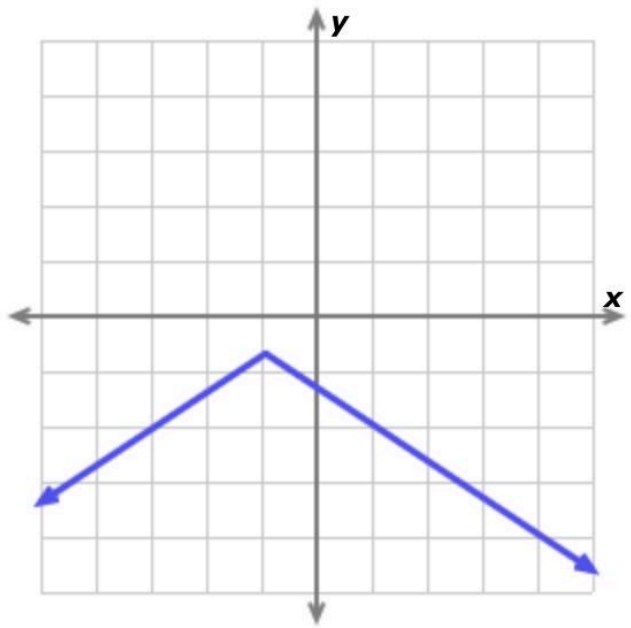
Skip to next question >

Which of these relations are functions? Select all that apply.









Save answer

Find the slope of the line $y = -9x - 9$.

Write your answer as an integer or as a simplified proper or improper fraction.

Save answer



What is the domain of this function?

(−10, 3)
(3, 2)
(0, −8)
(8, −4)
(7, 6)

{2, 3, 4, 6, 8}

{−8, −4, 2, 3, 6}

{0, 3, 7, 8}

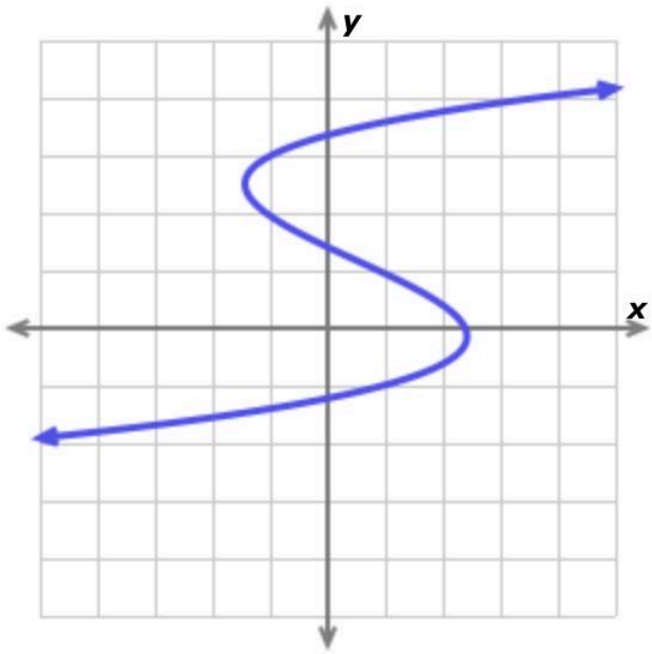
{−10, 0, 3, 7, 8}

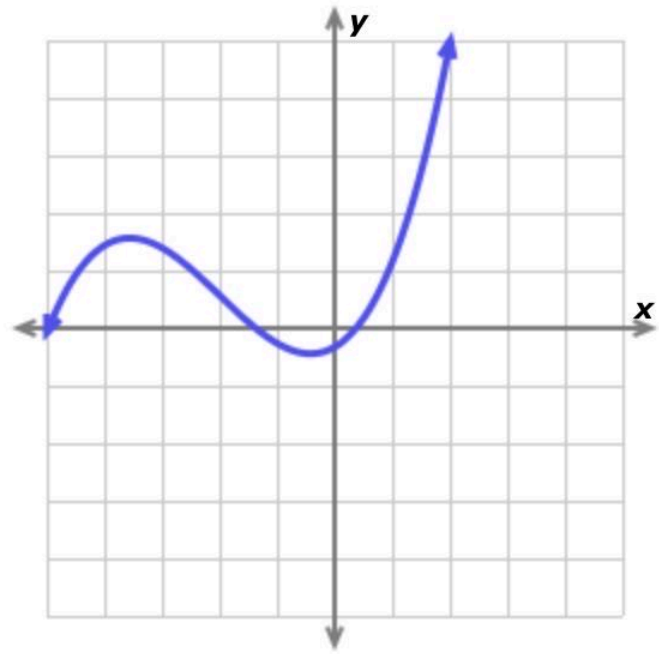
Save answer

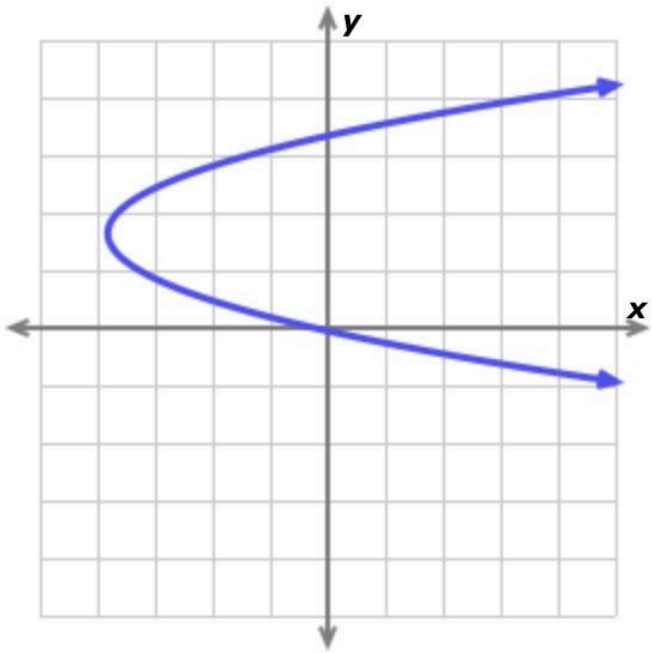


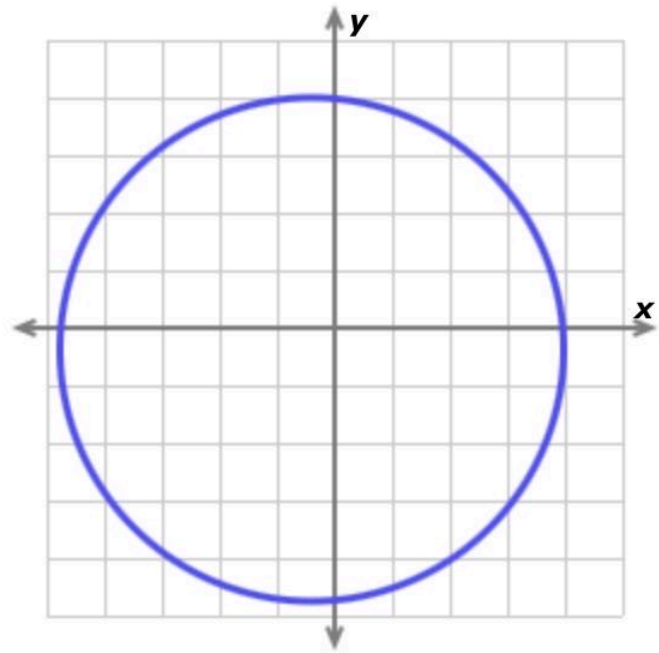
Skip to next questior

Which of these relations is a function?



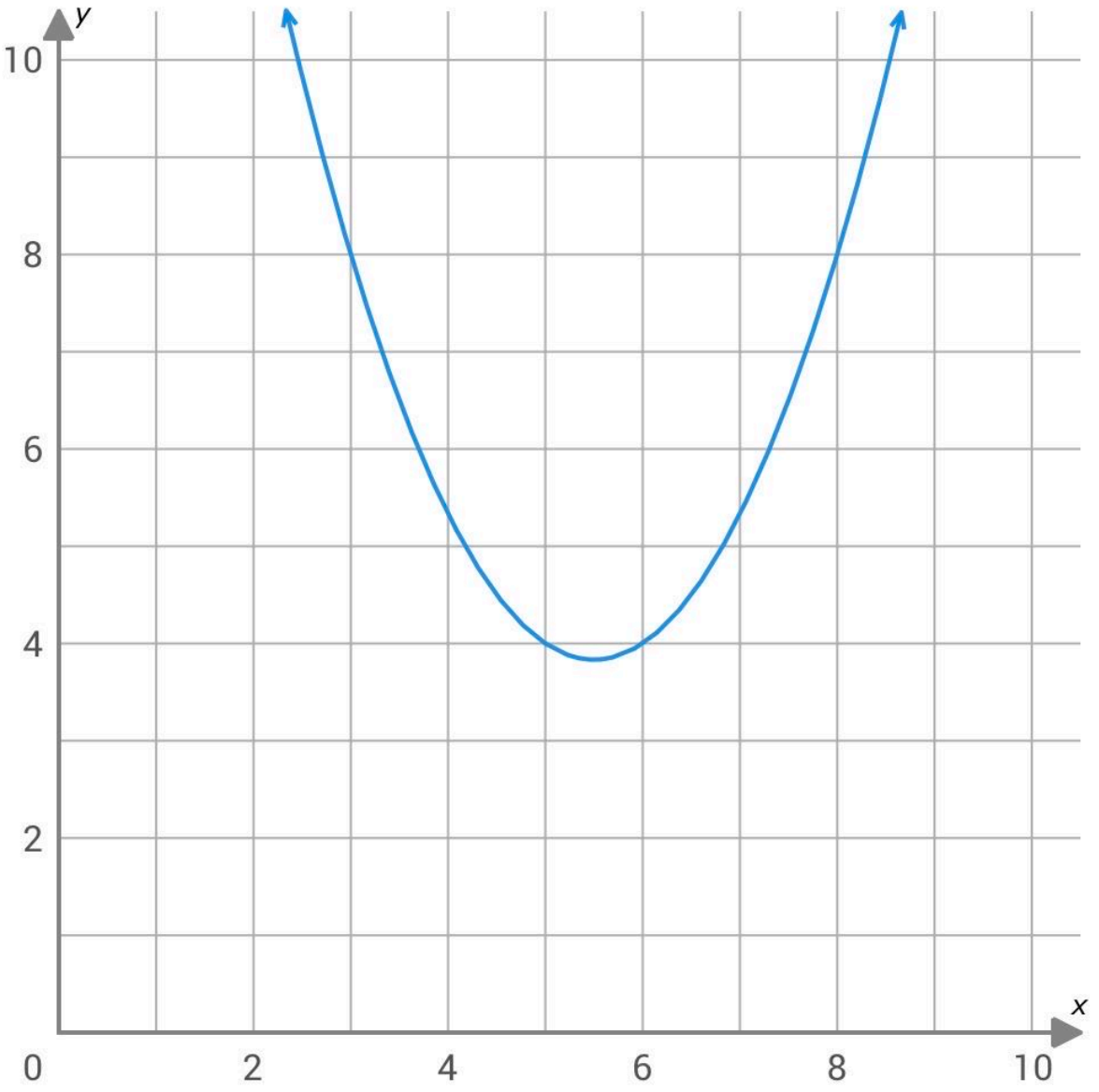






Save answer

Use the graph of the function to complete the table.



input	output
3	8
6	
8	

Save answer

What is the domain of this function?

(9, −2)
(6, −10)
(−3, 9)
(5, 2)

{−6, −3, 5, 9}

{−10, −2, 2, 9}

{−3, 5, 6, 9}

{2, 5, 6, 9}

Save answer



Skip to next question ›

What is the range of this function?

(8, −4)
(6, 10)
(0, 5)
(−8, 2)
(−1, −6)
(2, 5)

{−1, 0, 2, 6, 8}

{−4, 2, 5, 6, 10}

{−6, −4, 2, 5, 10}

{−8, −1, 2, 6, 8}

Save answer



Skip to next question

What is the range of this function?

(3, 7)
(4, 0)
(−3, 4)
(−1, 9)
(−10, −7)

{−10, −3, 1, 3, 4}

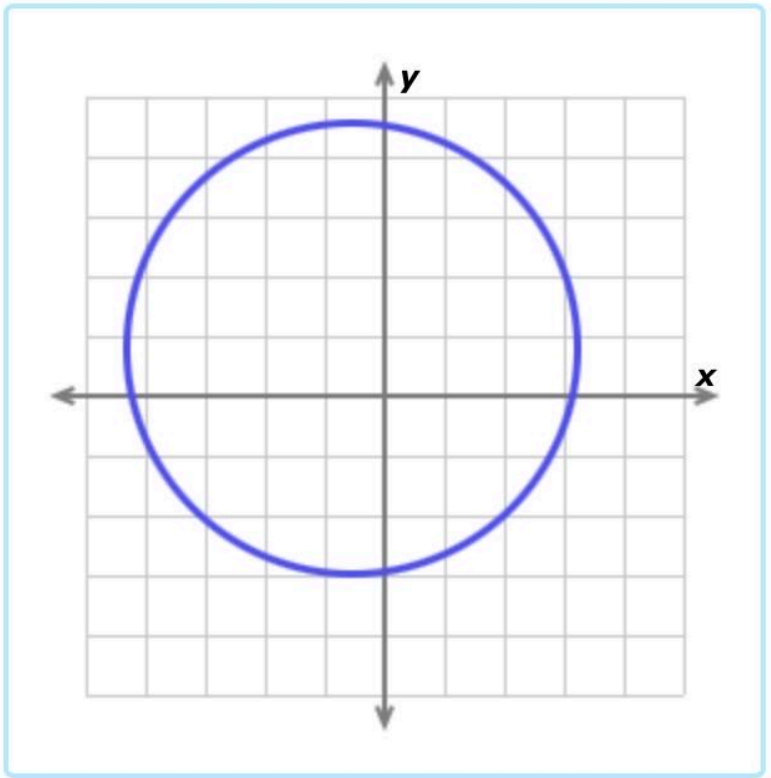
{−10, −3, −1, 3, 4}

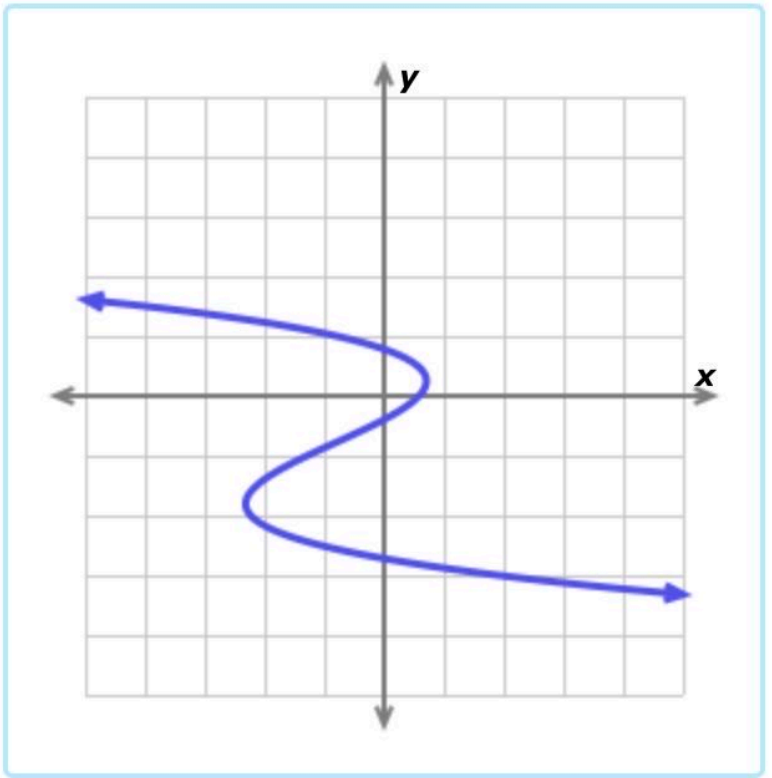
{−7, 0, 4, 7, 9}

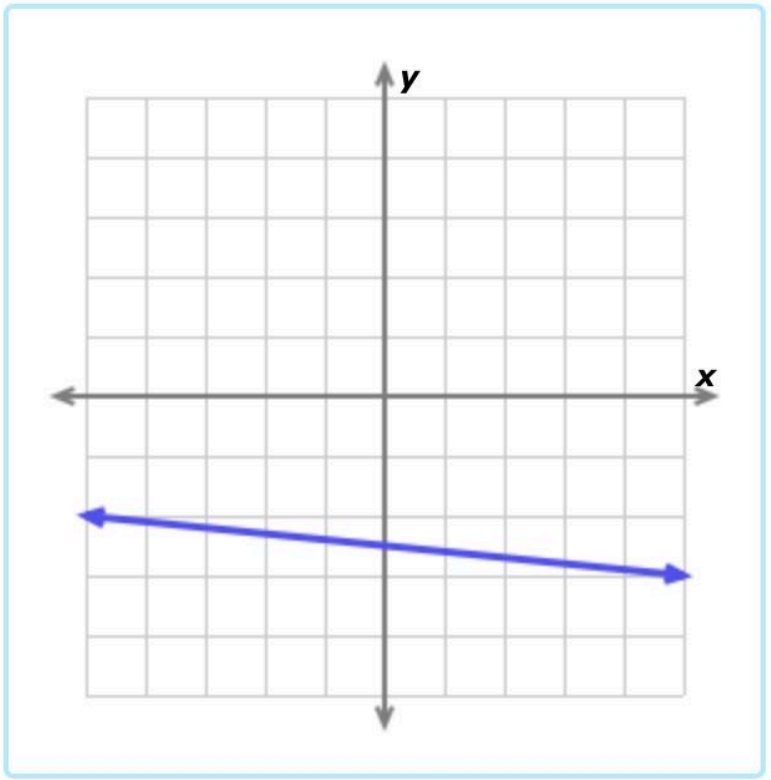
{−7, 4, 7, 9}

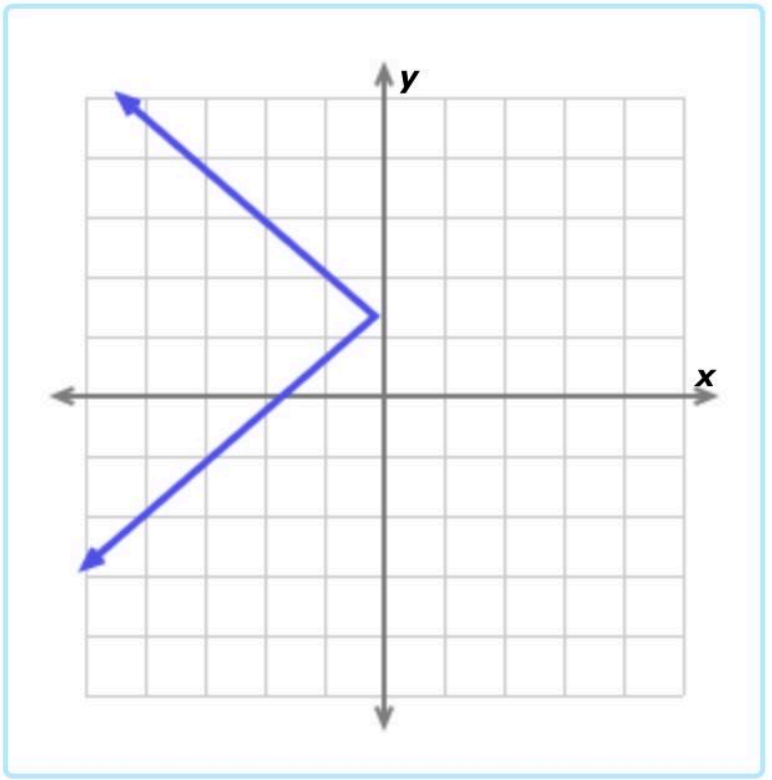
Save answer

Which of these relations is a function?

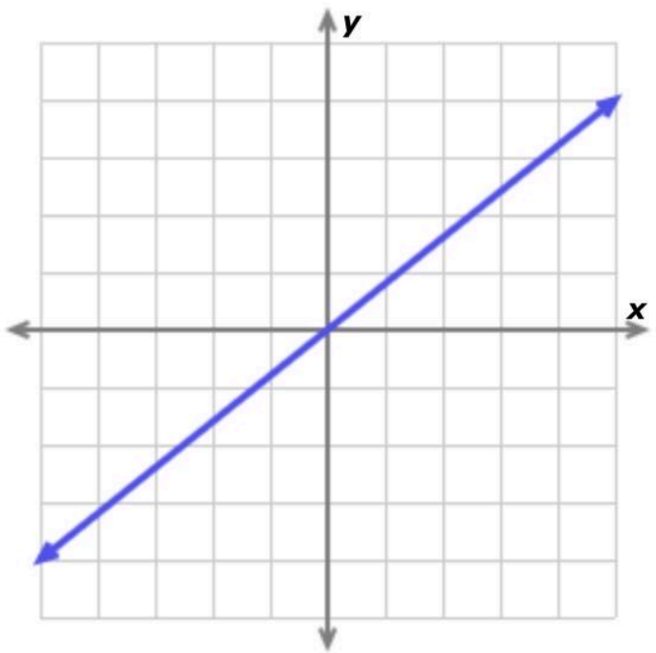








Look at this graph:



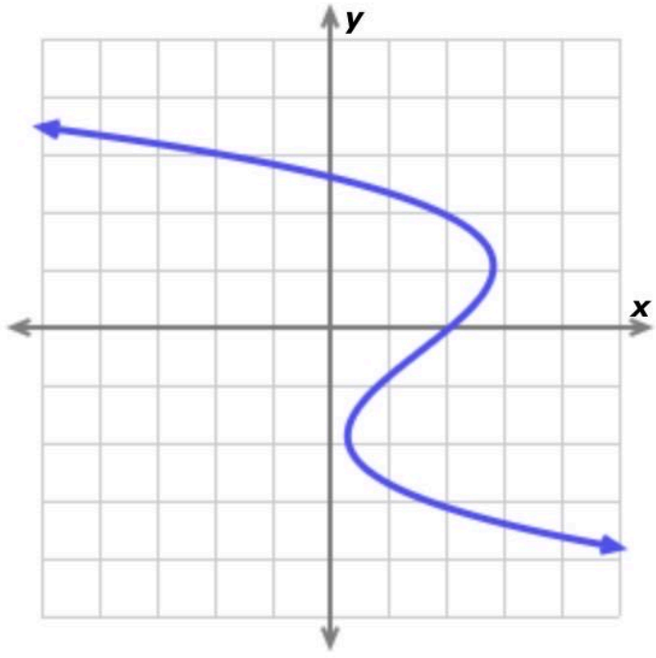
Is this relation a function?

yes

no

Save answer

Look at this graph:



Is this relation a function?

yes

no

Save answer

Look at this table:

x	y
15	1
5	17
6	16
1	8

Is this relation a function?

yes

no

Save answer

Look at this set of ordered pairs:

(5, 0)
(5, $^{-}2$)
(0, $^{-}8$)
(6, 20)

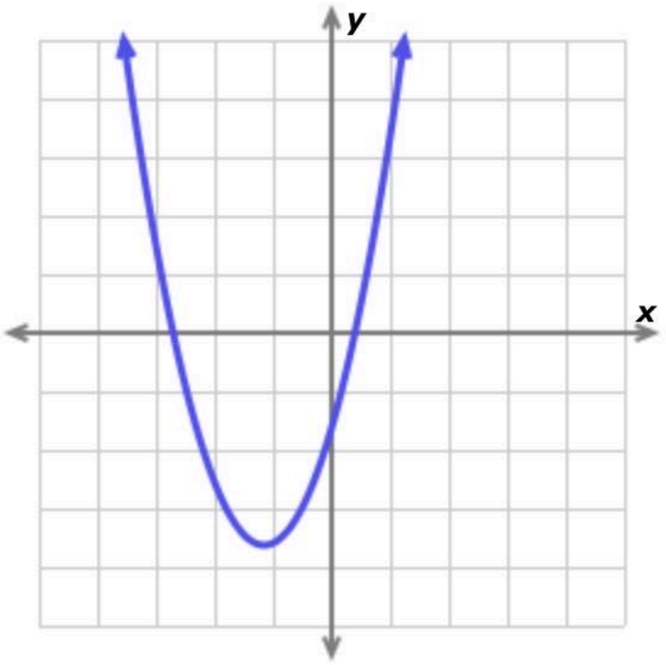
Is this relation a function?

yes

no

Save answer

Look at this graph:



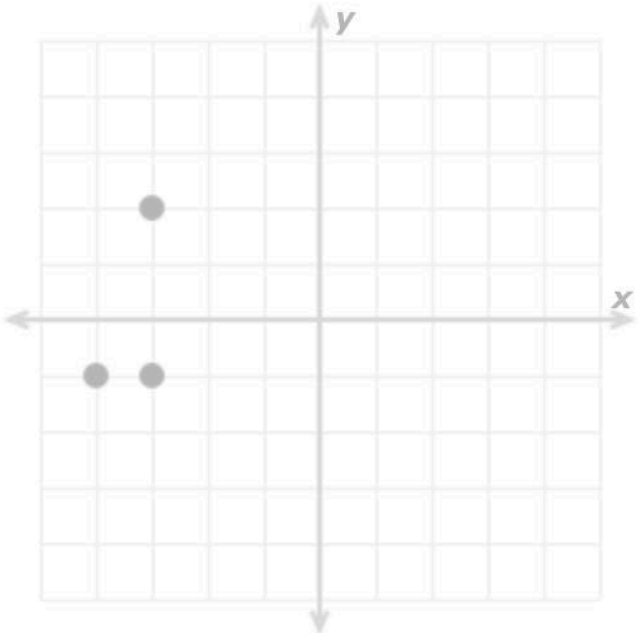
Is this relation a function?

yes

no

Save answer

Look at this graph:

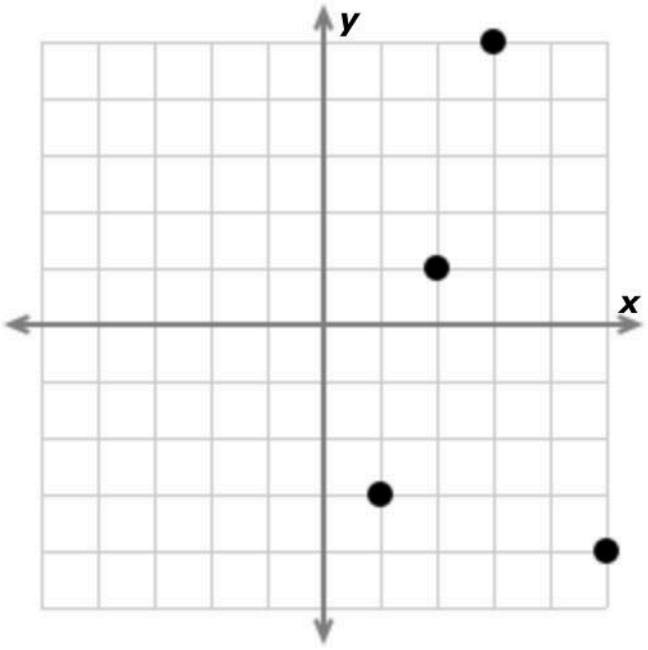


Is this relation a function?

yes

no

Look at this graph:

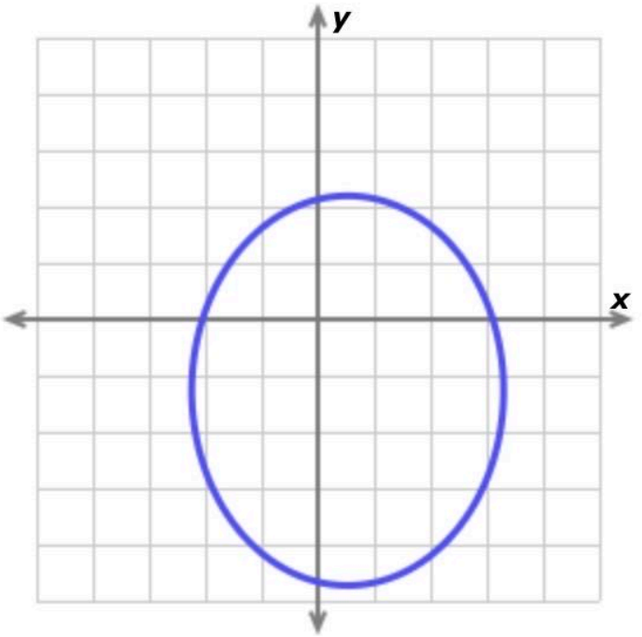


Is this relation a function?

yes

no

Look at this graph:



Is this relation a function?

yes

no

Save answer

Look at this table:

x	y
0	-3
0	11
13	14

Is this relation a function?

yes

no

Look at this table:

x	y
13	3
15	-3
-4	20
11	3

Is this relation a function?

yes

no

Save answer

Look at this table:

x	y
-4	0
5	3
5	6
19	10

Is this relation a function?

yes

no

Look at this table:

x	y
9	8
3	6
6	4

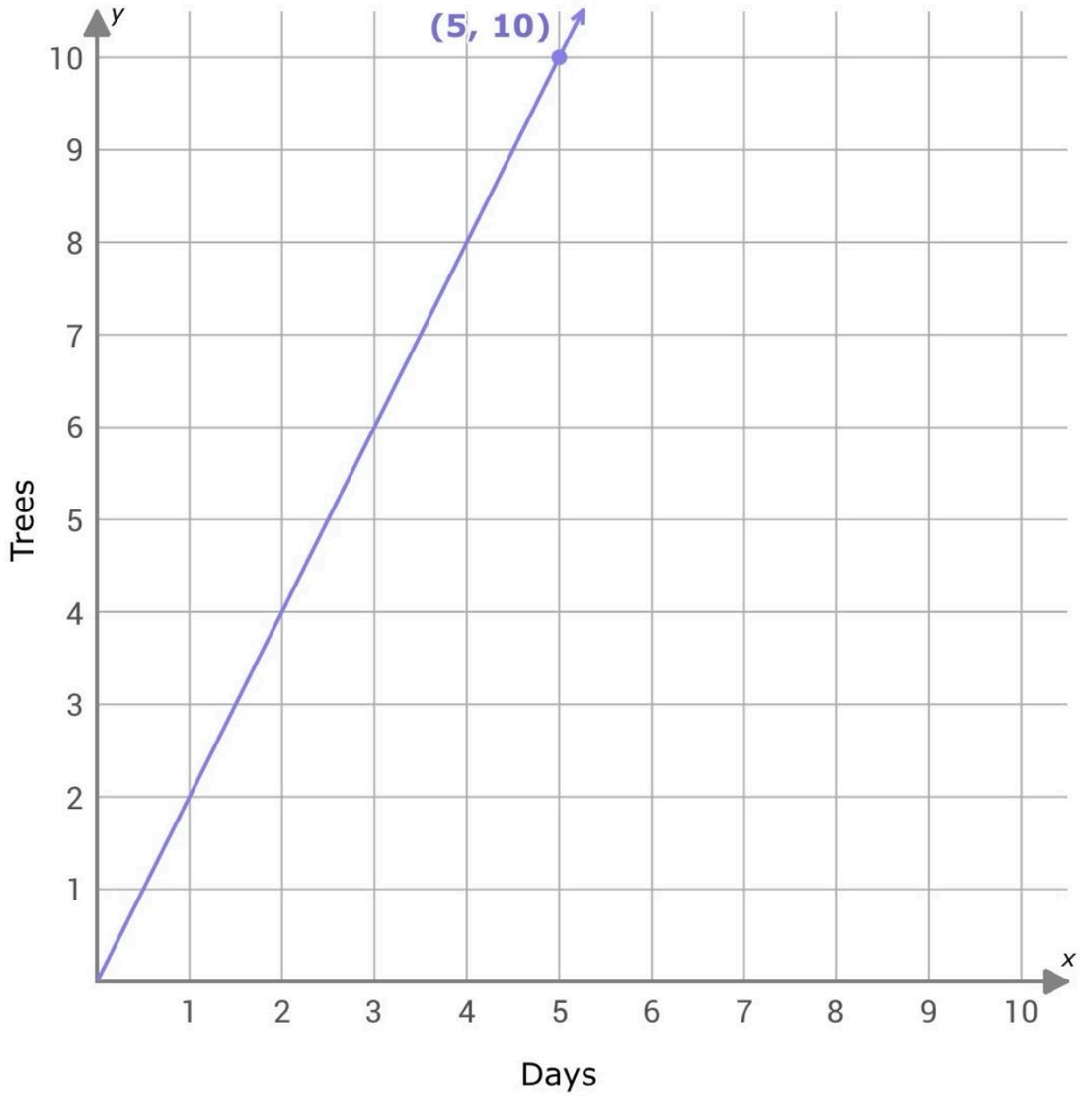
Is this relation a function?

yes

no

After a shipwreck, Emmet is stranded on a desert island. He needs to build a shelter, so he makes an axe using materials from the wreckage and starts cutting down trees.

The graph shows the proportional relationship between the number of days Emmet is stranded on the island, x, and the total number of trees he cuts down, y.



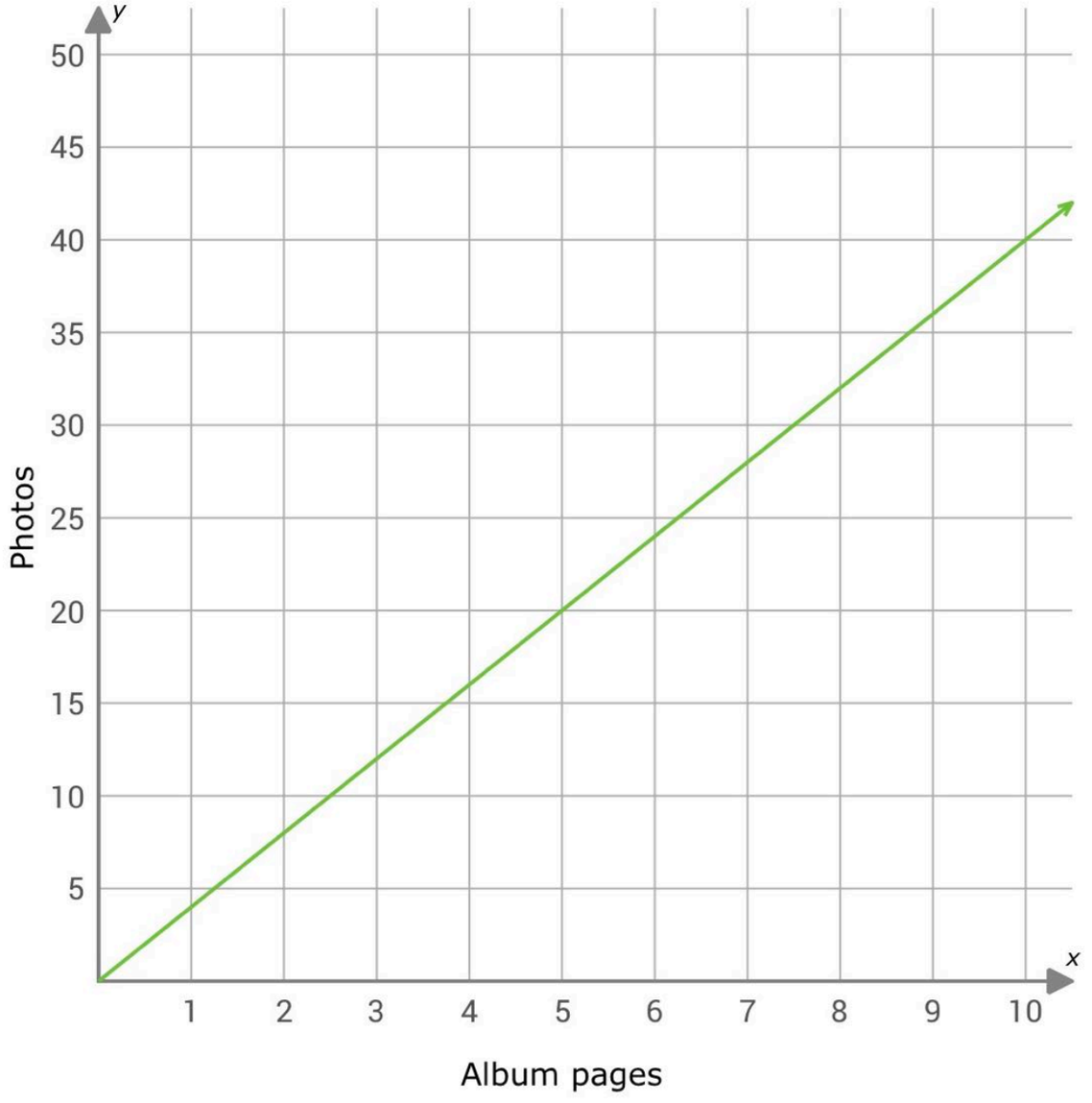
What does the point (5, 10) represent?

In his first 5 days on the island, Emmet cuts down a total of 10 trees.

In his first 10 days on the island, Emmet cuts down a total of 5 trees.

This summer, Antonio fulfilled a lifelong dream and traveled to Kenya. He took lots of photos and wants to put them all in his new photo album binder. First, he needs to buy album pages.

This graph shows the relationship between the number of album pages Antonio buys, x, and the number of photos the album will hold, y.



Do x and y have a proportional relationship?

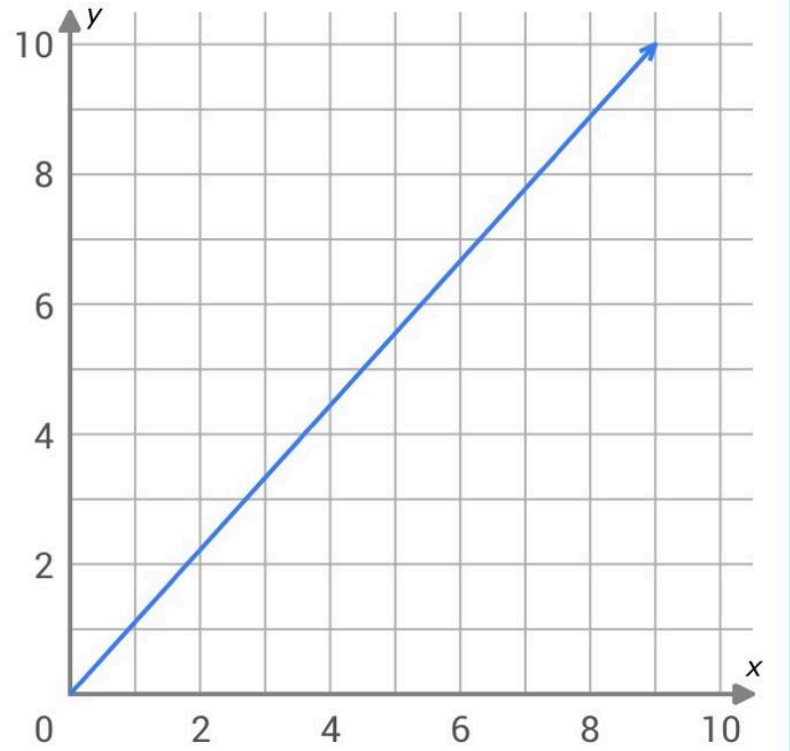
- yes
- no

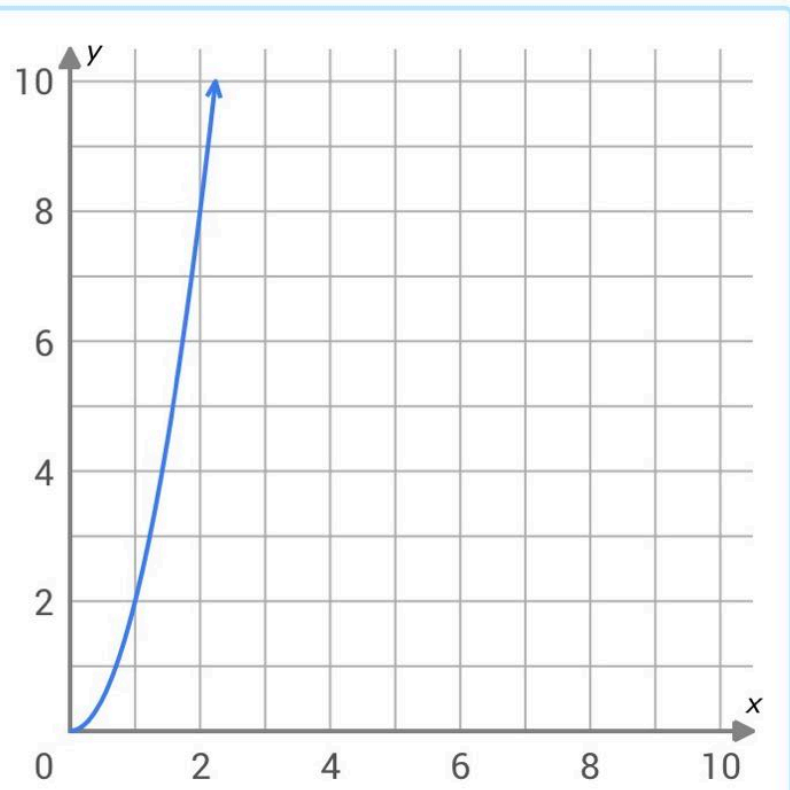
Select the tables that show a proportional relationship between x and y.

x	y
2	5
6	7
8	8

x	y
6	6
8	8
10	10

Select the graphs that show a proportional relationship between x and y.





Select the tables that show a proportional relationship between x and y.

x	y
3	3
6	7
15	19

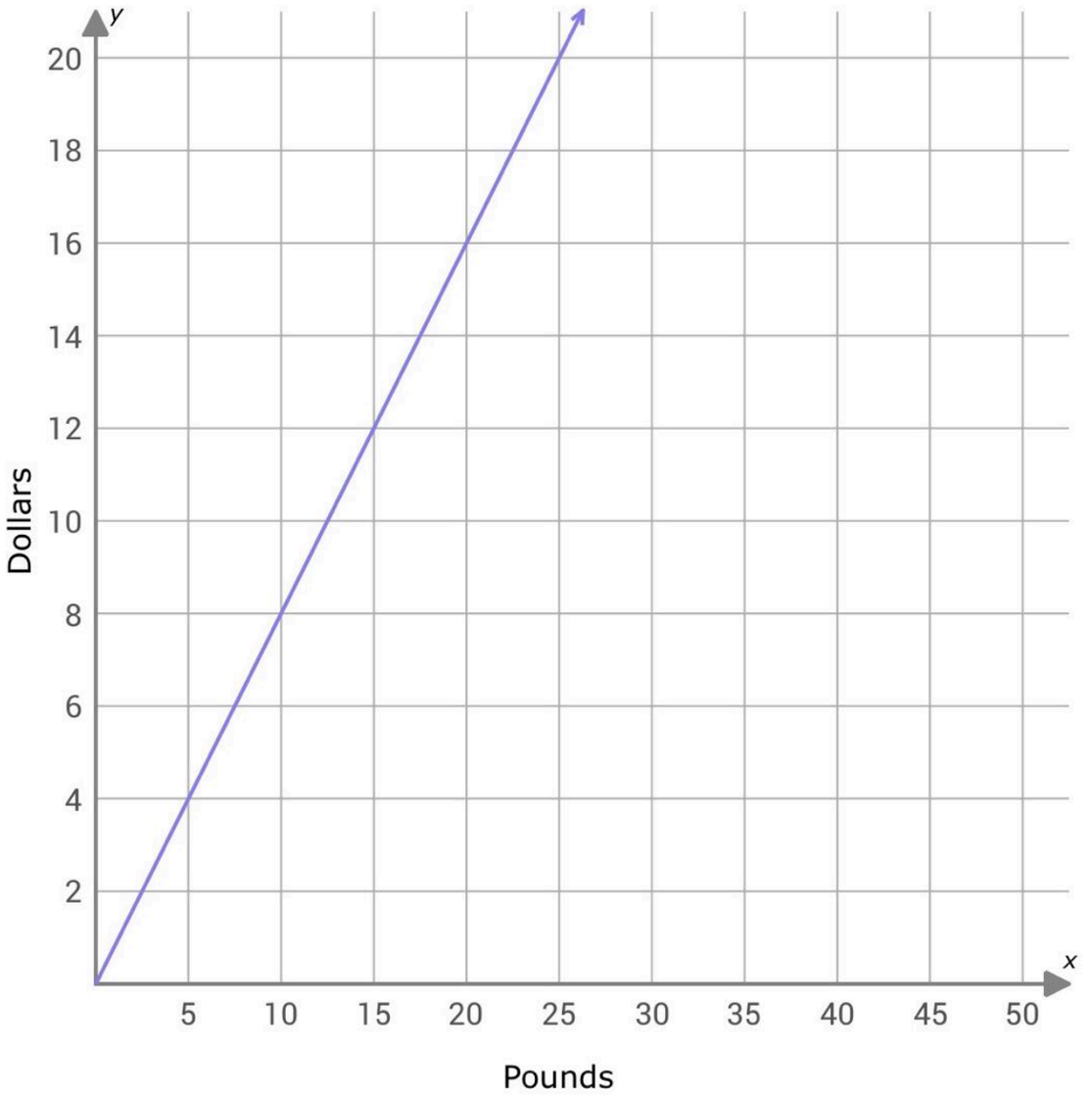
x	y
2	5
4	10
8	20

Save answer



Marshall is the executive chef at a Mediterranean restaurant famous for its spicy tomato sauce. Every week, he takes inventory and then places an order for what he needs. Since his spicy tomato sauce is so popular, Marshall needs to order more tomatoes every week.

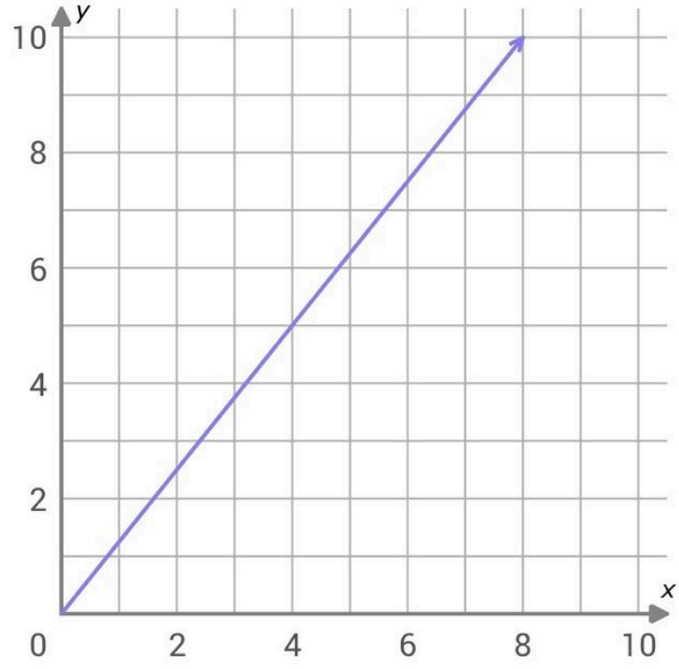
This graph shows the relationship between the amount of tomatoes he orders (in pounds), x, and their cost (in dollars), y.

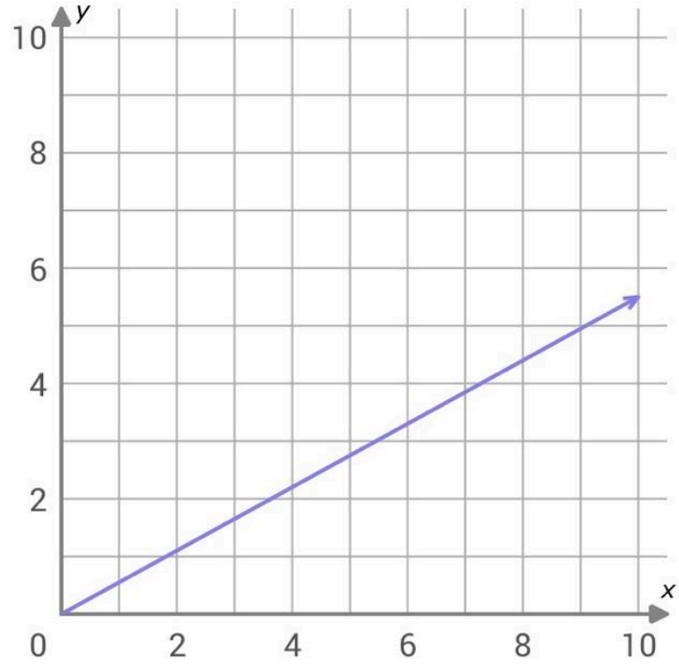


Do x and y have a proportional relationship?

- yes
- no

Select the graphs that show a proportional relationship between x and y.





Select the equations that show a proportional relationship between x and y.

- [x] $y = 6x + 1$
- [x] $y = 4x$

Save answer

Select the equations that show a proportional relationship between x and y.

- [x] $y = 5x - 2$
- [x] $y = 10x$

Amy's space shuttle crew accidentally left her behind on a mysterious planet. She soon realizes that her body is aging much faster on this planet than it did on Earth.

This table shows the relationship between the number of weeks Amy spends on the planet, x, and the number of years she ages, y.

x (weeks)	y (years)
3	18
5	30
6	36
8	48

According to the values in the table, do x and y have a proportional relationship?

- yes
- no

How many solutions does this equation have?

$4(q - 5) = {-}20 + 4q$

no solution

one solution

infinitely many solutions

Save answer

How many solutions does this equation have?

$6z + 1 = 1 + 6z$

no solution

one solution

infinitely many solutions

Save answer

To make some extra money on the weekends, Maureen drives for the City Express ride sharing service. During regular hours, passengers are charged a flat fee plus an additional fee per mile of the ride.

This table shows the relationship between the distance (in miles) Maureen drives a passenger, x, and the amount (in dollars) the passenger is charged, y.

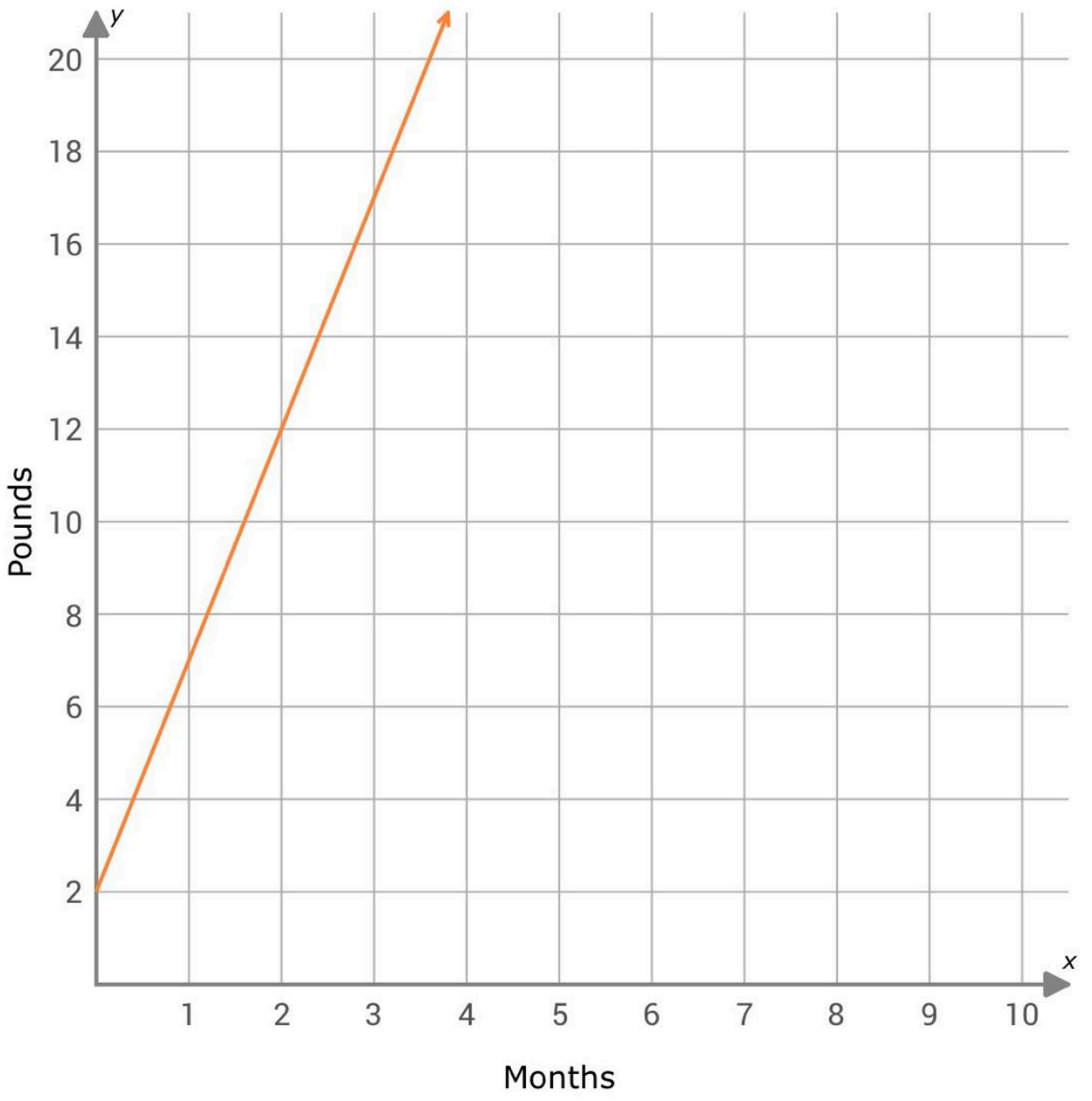
x (miles)	y (dollars)
5	$19
23	$73
28	$88
29	$91

According to the values in the table, do x and y have a proportional relationship?

- yes
- no

Alec's grandparents get him a puppy for his thirteenth birthday, and he names the puppy Lucky. Alec does a good job taking care of Lucky, and she's growing so fast!

This graph shows the relationship between Lucky's age (in months), x, and her weight (in pounds), y.



Do x and y have a proportional relationship?

- yes
- no

Ronnie wants to save up some money to buy a new smartphone, so he babysits on the weekends.

This table shows the relationship between the time Ronnie spends babysitting (in hours), x, and the amount of money he earns babysitting (in dollars), y.

x (hours)	y (dollars)
1	$11
2	$22
8	$88
9	$99

According to the values in the table, do x and y have a proportional relationship?

- yes
- no

Luther wants to get in shape, so he joins a local gym, Buff and Tough. For membership, the gym charges a registration fee plus monthly dues.

This table shows the relationship between the length of Luther's gym membership (in months), x, and the total cost of his membership (in dollars), y.

x (months)	y (dollars)
2	$49
3	$65
4	$81
5	$97

According to the values in the table, do x and y have a proportional relationship?

- yes
- no

The housekeeping staff at the Weary Traveler Hotel rotates job assignments every day. Today, Lacey is in charge of folding the freshly cleaned bedsheets.

This table shows the relationship between the amount of time (in minutes) Lacey spends folding, x, and the number of bedsheets she folds, y.

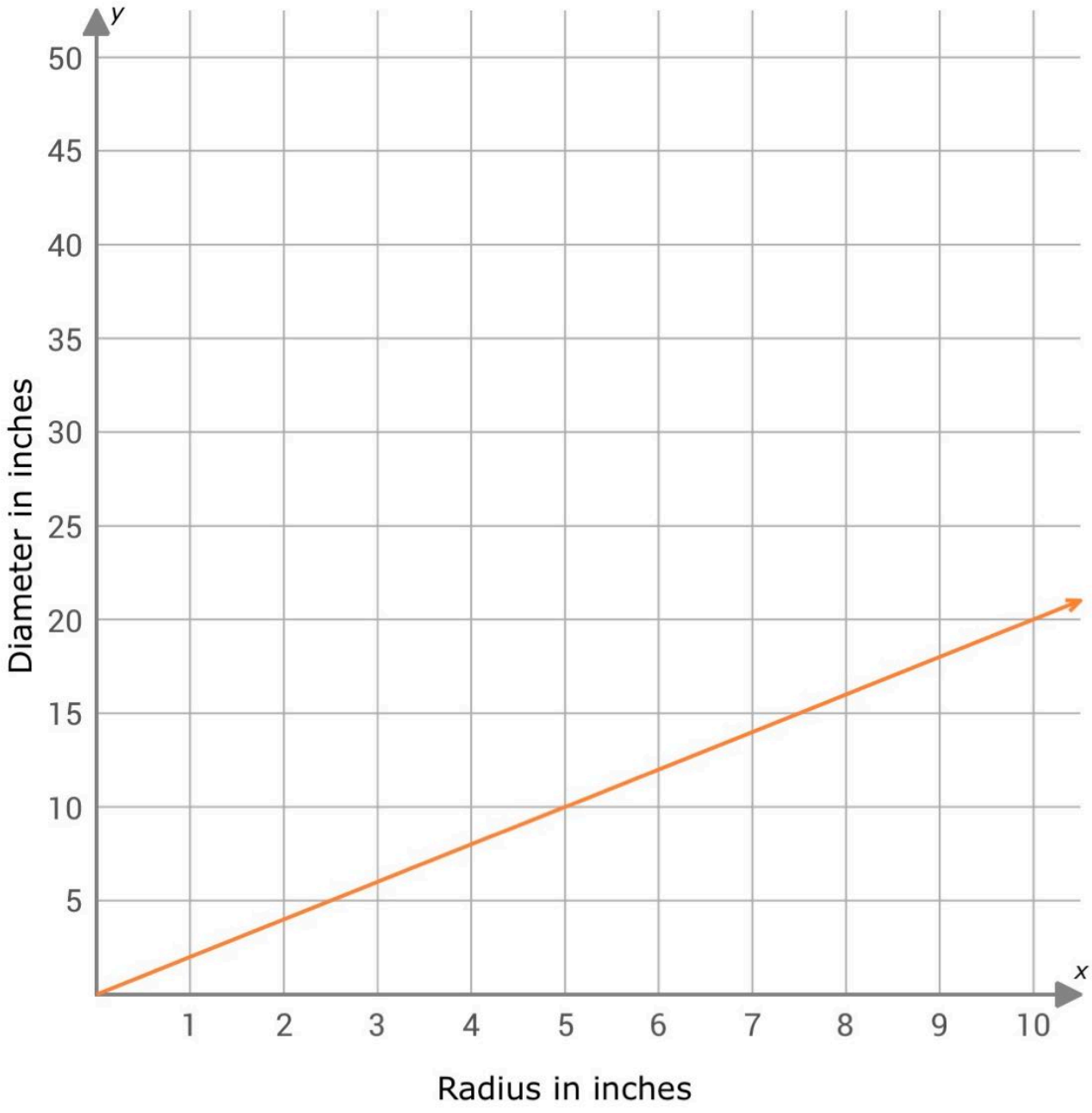
x (minutes)	y (bedsheets)
7	35
9	45
10	50
17	85

According to the values in the table, do x and y have a proportional relationship?

- yes
- no

Roxanne is measuring the diameter and the radius of different-sized circles.

This graph shows the relationship between the radius of the circle in inches, x, and the diameter of the circle in inches, y.



Do x and y have a proportional relationship?

- yes
- no

Lorenzo loves to read and discuss books, so he joined a book club. The first book he read as a member of the club was a novel set in ancient Rome.

This table shows the relationship between the time Lorenzo has spent reading the novel (in hours), x, and the number of pages he has read, y.

x (hours)	y (pages)
1	20
3	60
4	80
5	100

According to the values in the table, do x and y have a proportional relationship?

- yes
- no

How many solutions does this equation have?

$10k = 10k + 8$

no solution

one solution

infinitely many solutions

Save answer

How many solutions does this equation have?

$-7f = 8 - 8f$

no solution

one solution

infinitely many solutions

Save answer

How many solutions does this equation have?

$-5 + 10j = 2 + 10j$

- no solution
- one solution
- infinitely many solutions

Save answer

How many solutions does this equation have?

$6 - 9r = {}^{-}7r$

- no solution
- one solution
- infinitely many solutions

Save answer

How many solutions does this equation have?

$0 = 10r - 10r$

no solution

one solution

infinitely many solutions

Save answer